29 Connections, Signals and Switches

29.1 Main and Extension COP

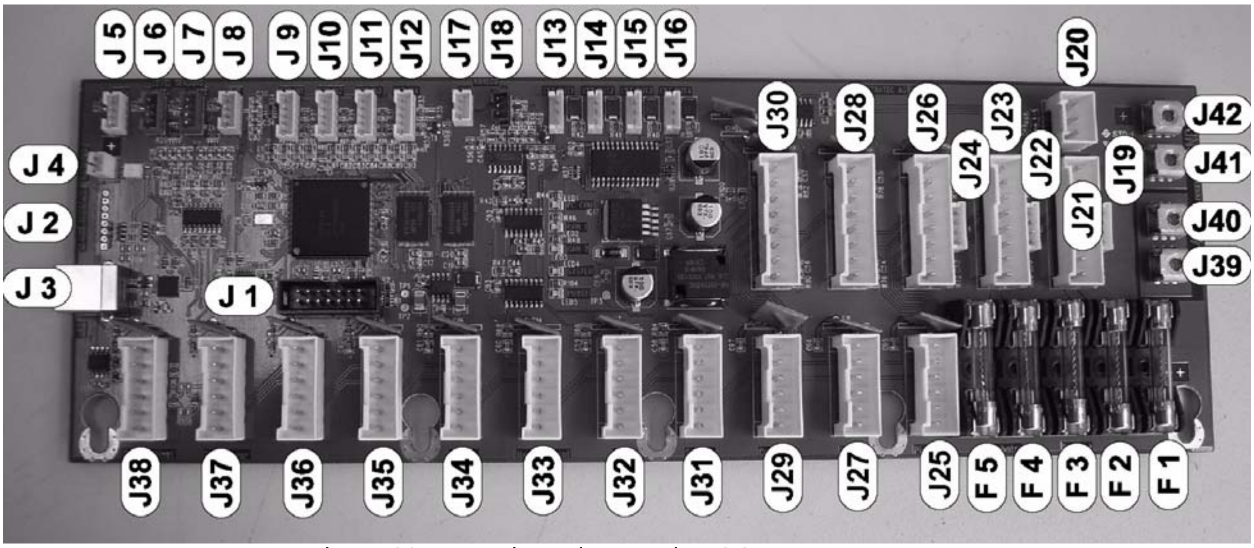

Figure 29-1: Main and Extension COP

29.1.1 COP Main

Slot No.	Description	Slot No.	Description
J 1	Download Monitor	J 22	Not used
J 2	To USB Interface	J 23	Cooling Module
J 3	USB	J 24	Not used
J 4	Shutdown	J 25	Not used
J 5	Level Sensor Wash	J 26	Pipettor
J 6	Level Sensor System	J 27	Not used
J 7	Level Sensor Waste	J 28	TCR Loading
J 8	Level Sensor Extension	J 29	Not used
J 9	Ext In 1	J 30	PWR-Ch.1: Alitea PWR-Ch.2: LCMP Pump PWR-Ch.3: Diener Pump
J 10	Ext In 2	J 31	Sample
J 11	Ext In 3	J 32	Reagent
J 12	Ext In 4	J 33	Mag Wash 1
J 13	Ext Out 1 (Wash Valve)	J 34	Mag Wash 2
J 14	Ext Out 2 (DI Water Valve)	J 35	Frontcover
J 15	Ext Out 3 (empty)	J 36	Univers. Fluid RFID
J 16	Ext Out 4 (Beeper)	J 37	not used (Termination)
J 17	Analog Out	J 38	COP Extension Out
J 18	Analog In	J 39	24V Main Power In
J 19	Vacuum Pump	J 40	24V Main Power In
J 20	Not used	J 41	GND
J 21	Cooling Module	J 42	GND

Table 29-1: COP Main

29.1.2 COP Extension

Slot No.	Description	Slot No.	Description
J 1	Download Monitor	J 22	Not used
J 2	To USB Interface	J 23	Incubator 63°C
J 3	USB	J 24	Not used
J 4	Shutdown	J 25	Not used
J 5	Level Sensor Wash	J 26	Incubator 43,7°C
J 6	Level Sensor System	J 27	Not used
J 7	Level Sensor Waste	J 28	Incubator 42,7°C
J 8	Level Sensor Extension	J 29	Not used
J 9	Ext In 1	J 30	PWR-Ch.1: AMP Load Station PWR-Ch.2: HPA Load Station
J 10	Ext In 2	J 31	2x VP9101 / Y-Cable
J 11	Ext In 3	J 32	Sample Load
J 12	Ext In 4	J 33	Output Queue
J 13	Ext Out 1 (Wash Valve)	J 34	Luminometer
J 14	Ext Out 2 (DI Water Valve)	J 35	Input Queue
J 15	Ext Out 3 (empty)	J 36	COP Extension In
J 16	Ext Out 4 (Beeper)	J 37	Not used
J 17	Analog Out	J 38	Not used
J 18	Analog In	J 39	24V Main Power In
J 19	Not used	J 40	24V Main Power In
J 20	Not used	J 41	GND
J 21	Linear Distributor	J 42	GND

Table 29-2: COP Extension

29.2 Front Module Cover

29.2.1 Sample-TCR Display Board

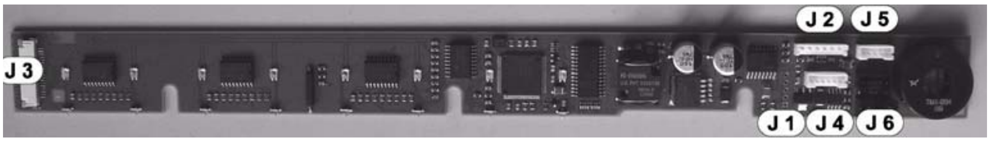

Figure 29-2: Sample TCR Display Board

Slot No.	Description
J 1	Download Monitor
J 2	MAIN IN Power / CAN
J 3	Reagent + Diti Display Board
J 4	Interlock
J 5	Top Cover 1
J 6	Top Cover 2

Table 29-3: Sample Display Board

29.2.2 Reagent Display Board

Figure 29-3: Reagent Display Board

Slot No.	Description
J 1	From Diti Display Board

Table 29-4: Reagent Display Board

29.2.3 DITI Display Board

Figure 29-4: DITI Display Board

Slot No.	Description
J 1	From Sample-TCR Display CU Board
J 2	To Reagent Display Board

Table 29-5: Diti Display Board

29.3 Pipettor

29.3.1 PIP CU Board

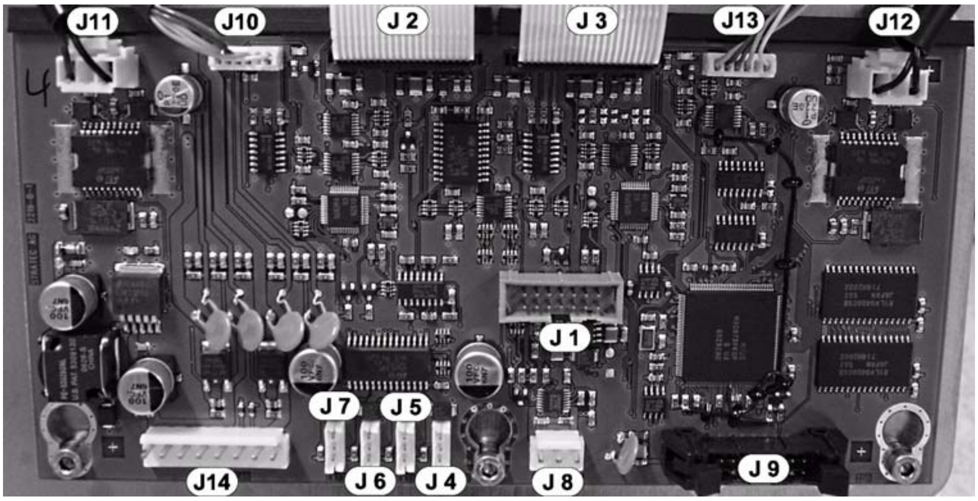

Figure 29-5: PIP CU Board

Slot No.	Description
J 1	Download Monitor
J 2	Left Arm FFC Connector
J 3	Right Arm FFC Connector
J 4	PIP Rinse Pump
J 5	EXT 1
J 6	EXT 2
J 7	EXT 3
J 8	Waste Pump
J 9	Syringe Pump
J 10	X-Encoder-L
J 11	X-Motor-L
J 12	X-Motor-R
J 13	X-Encoder-R
J 14	Main PWR CAN

Table 29-6: PIP CU Board

29.3.2 X-Sledge 2 Board

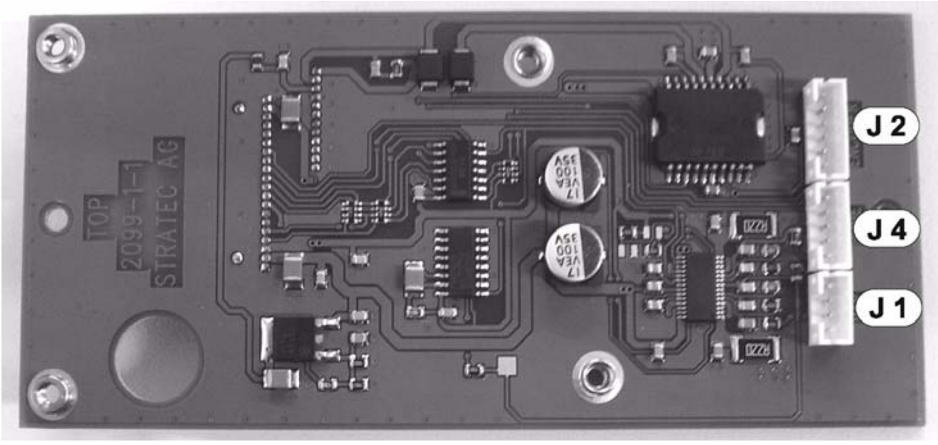

Figure 29-6: X-Sledge 2 Board, front view

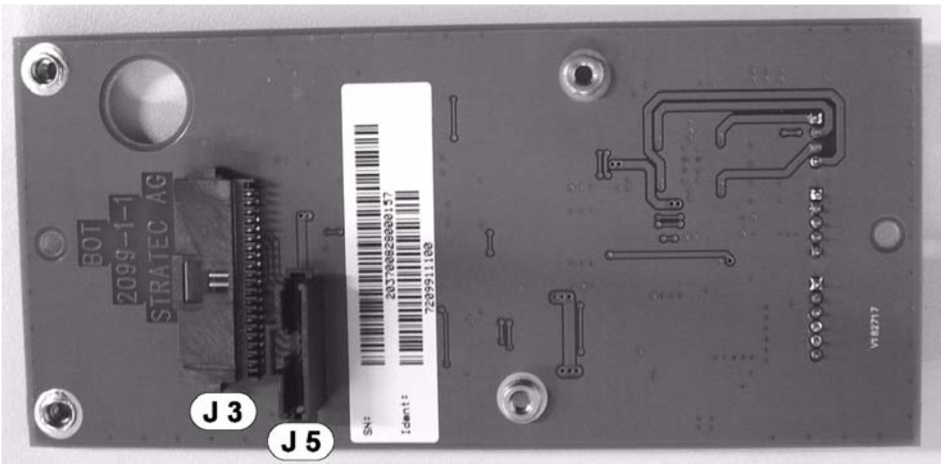

Figure 29-7: X-Sledge 2 Board, rear view

Slot No.	Description
J 1	Y-Motor
J 2	Y-Encoder
J 3	FFC Connector, Main Board
J 4	X-Home
J 5	FCC Connector, Y-Z Sledge

Table 29-7: X-Sledge 2 Board

29.3.3 Y-Sledge ZSP Board

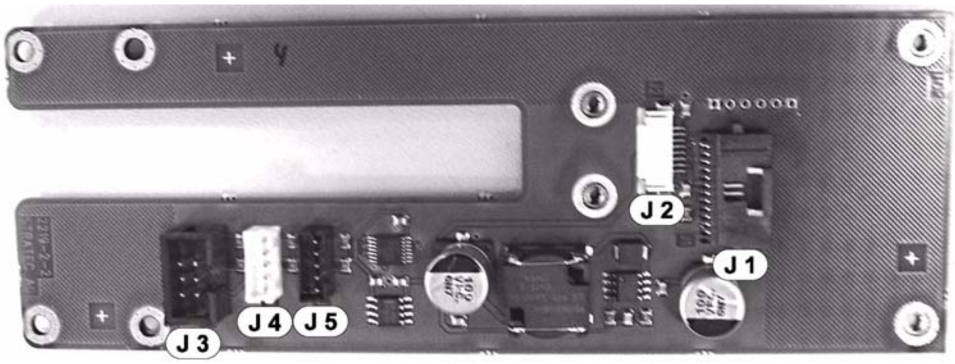

Figure 29-8: Y-Sledge ZSP Board

Slot No.	Description
J 1	X-Sledge
J 2	ZS-PIP
J 3	Z-Motor
J 4	Z-Home
J 5	Y-Home

Table 29-8: Y-Sledge ZSP Board

29.3.4 ZS-PIP CU Board

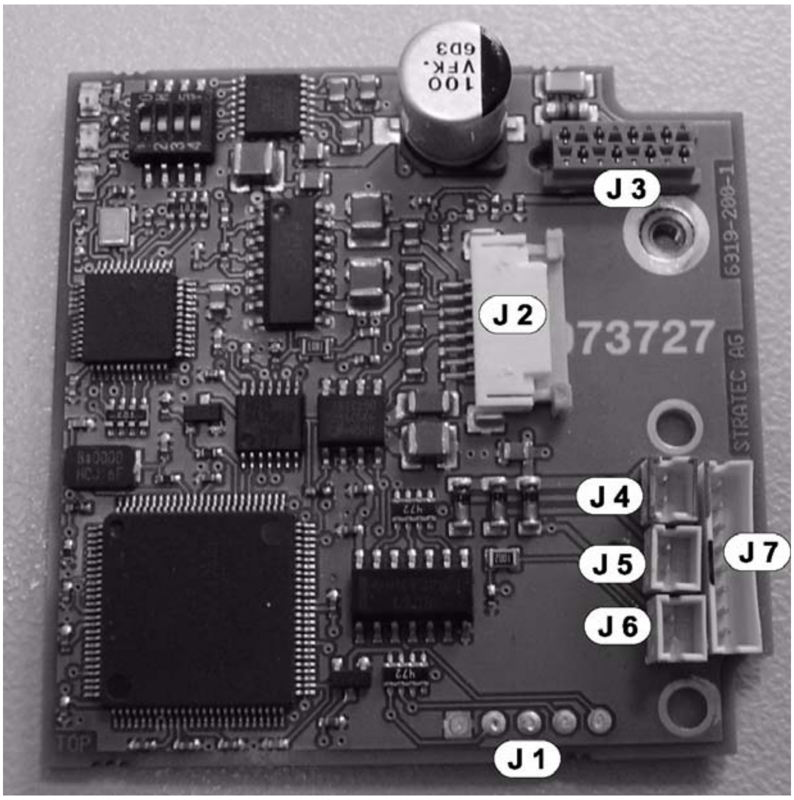

Figure 29-9: ZS-Pip CU Board

Slot No.	**Description**
J 1	Download Monitor
J 2	Main in Power / CAN
J 3	P-Motor
J 4	Valve 1
J 5	Valve 2
J 6	Valve 3
J 7	AP Head Connector

Table 29-9: ZS -PIP CU Board

29.3.5 AP HEAD Board

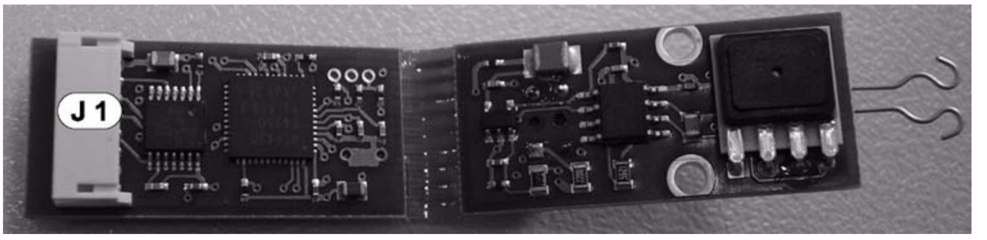

Figure 29-10: AP Head Board

Slot No.	Description
J 1	ZS PIP CU Board

Table 29-10: AP Head Board

29.4 Reagent Bay

29.4.1 Reagent CU Board

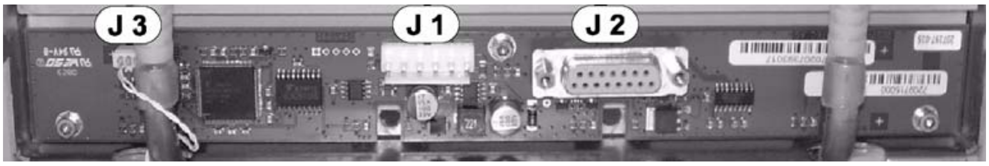

Figure 29-11: Reagent CU Board

Slot No.	Description
J 1	Power-Can
J 2	Scanner
J 3	Temperature Sensor

Table 29-11: Reagent CU Board

29.5 Sample Bay

29.5.1 Sample CU Board

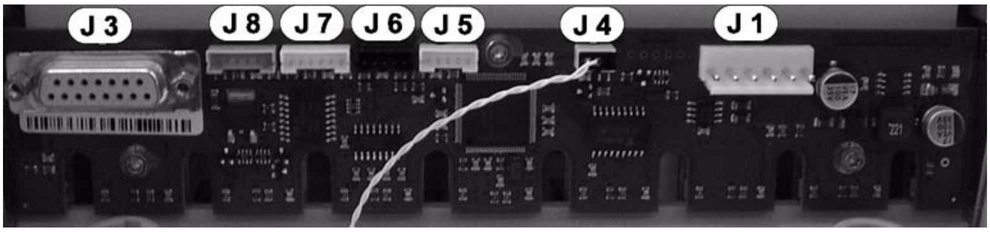

Figure 29-12: SAMPLE CU Board

Slot No.	Description
J 1	Power-Can
J 3	Scanner Connector
J 4	Temperature Sensor
J 5	Drawer Sens. 1
J 6	Drawer Sens. 2
J 7	Interlock 1
J 8	Interlock 2

Table 29-12: Overview Sample Bay

29.6 TCR Carousel

29.6.1 TCR Carousel CU Board

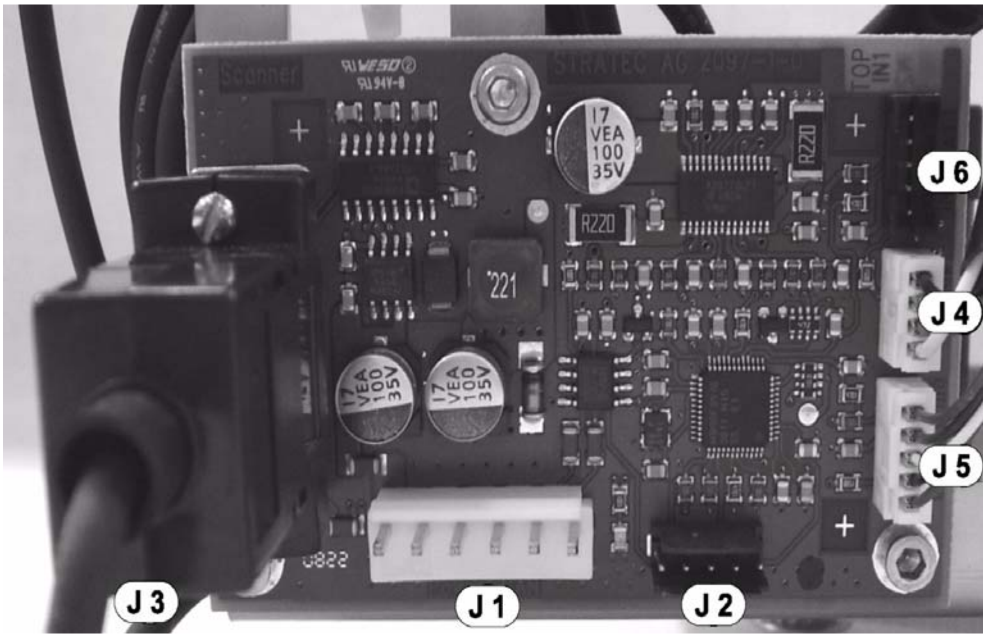

Figure 29-13: Carousel CU Board

Slot No.	Description
J 1	Power-Can
J 2	Download
J 3	Scanner Connector
J 4	Motor
J 5	MTR-Home
J 6	Reserve

Table 29-13: Carousel CU Board

29.7 Distributor

29.7.1 Distributor CU Board

This section contains a total overview and several sectional overviews of the electric components of the distributor. All electric connections, signal lamps and switches are described.

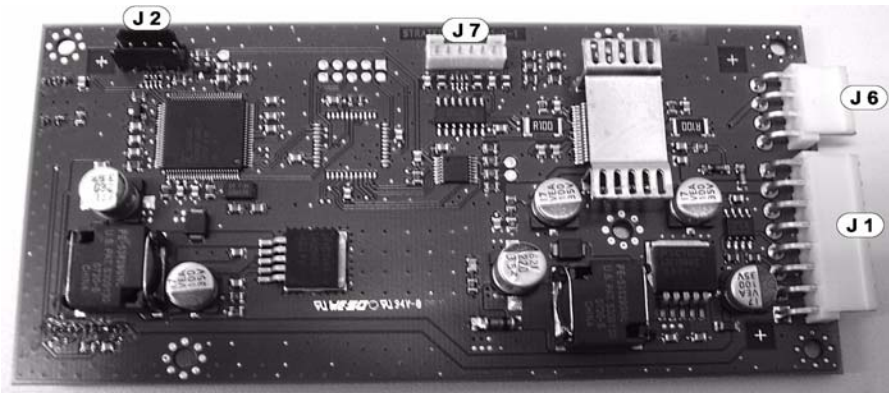

Figure 29-14: Distributor CU Board, front view

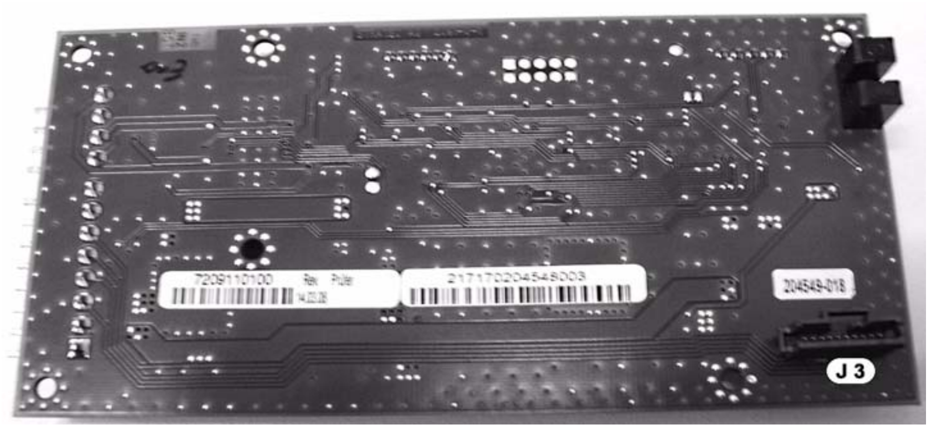

Figure 29-15: Distributor CU Board, rear view

Slot No.	Description
J 1	Main-Input
J 2	Download
J 3	Sledge CU Board
J 6	X-Drive
J 7	X-Drv Enc

Table 29-14: Distributor CU Board

29.7.2 Sledge CU Board

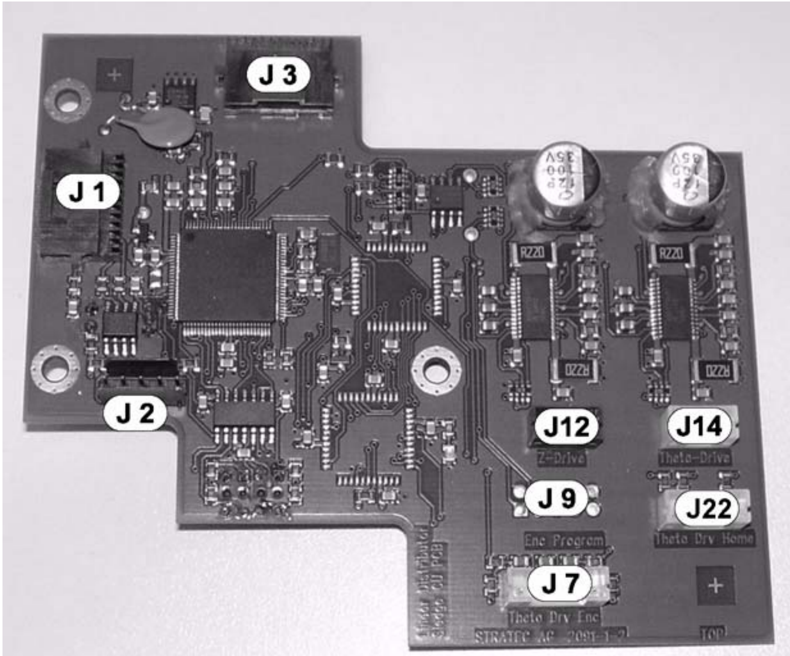

Figure 29-16: Linear Distributor Sledge CU Board

Slot No.	Description
J 1	MAIN CU Board
J 2	Monitor
J 3	Hook CU Board
J 7	Theta Drv Enc
J 9	Enc Program
J 12	Z-Drive
J 14	Theta-Drive
J 22	Theta Drv Home

Table 29-15: Linear Distributor Sledge CU Board

29.7.3 Hook CU Board

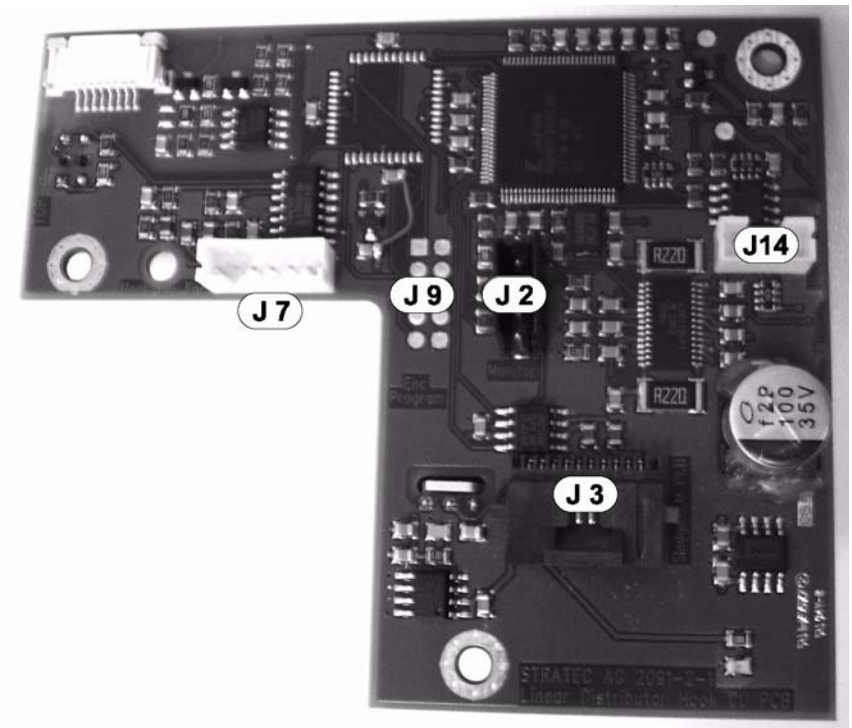

Figure 29-17: Distributor Hook CU Board

Slot No.	Description
J 1	Not used
J 2	Monitor
J 3	Sledge CU Board
J 7	Hook Drive Encoder
J 9	Enc Program (not used)
J 14	Hook-Drive

Table 29-16: Distributor Hook CU Board

29.8 Load Station

29.8.1 Load Station CU Board

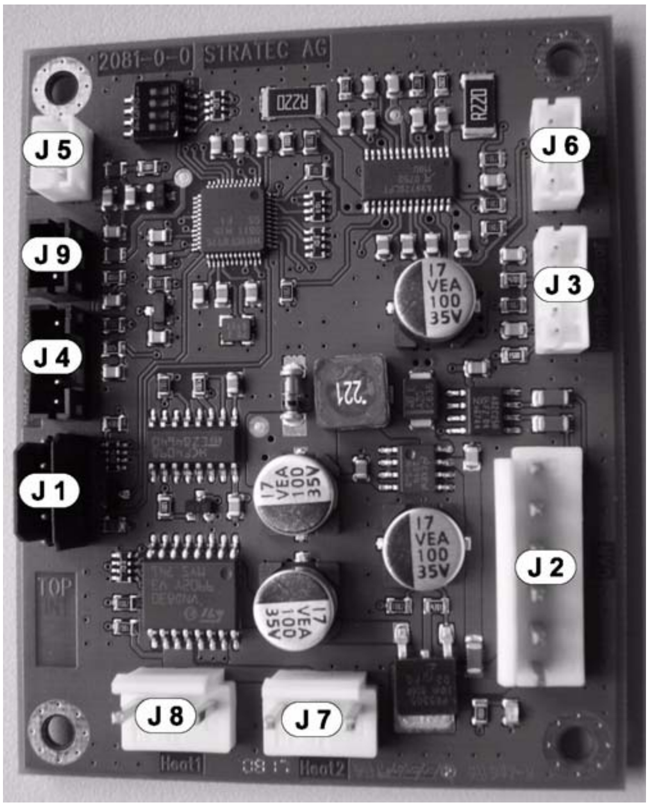

Figure 29-18: Load Station CU Board

Slot No.	Description
J 1	Download
J 2	CAN
J 3	Home Sensor
J 4	Reserve Sensor
J 5	Temp. 1
J 6	Motor
J 7	Heat 2
J 8	Heat 1
J 9	Temp. 2

Table 29-17: Load Station CU Board

29.9 Magnetic Wash

29.9.1 Mag Wash CU Board

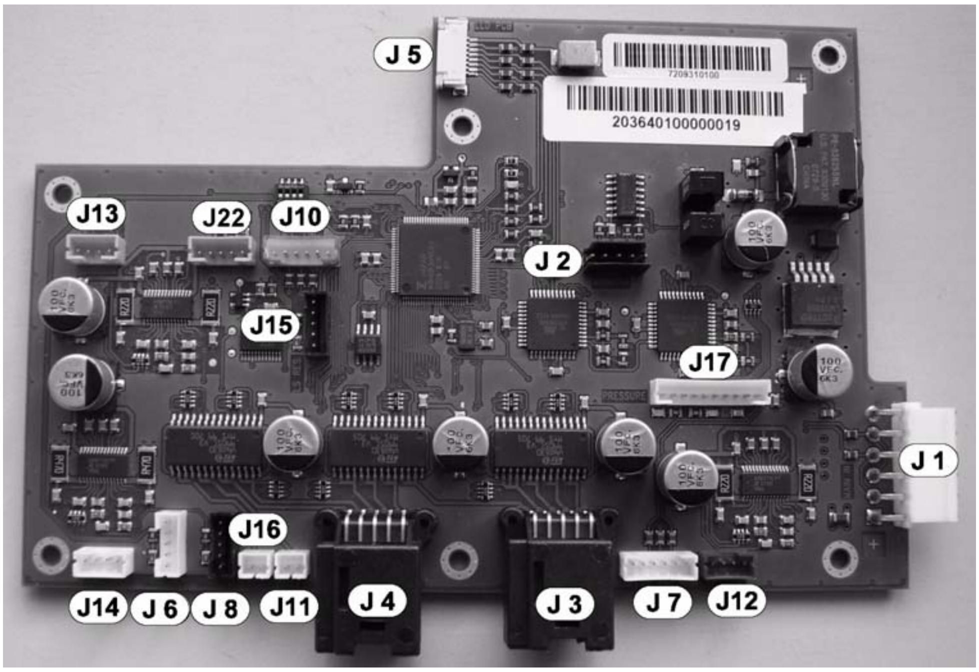

Figure 29-19: MAGWASH CU Board

Slot No.	Description
J 1	Main In
J 2	Monitor
J 3	Valve Outputs 6-10
J 4	Valve Outputs 1-5
J 5	Level Detection Board
J 6	Magnet Drive Home
J 7	ASPIRATOR DRIVE ENCODER
J 8	Magnet Drive End Of Travel
J 10	MIXER DRIVE ENCODER
J 11	Reserve Outputs 1+2
J 12	Aspirator Drive
J 13	Mixer Drive
J14	Magnet Drive
J 15	RESERVE (LS Res)
J 16	Reserve Outputs 1+2
J 17	Pressure
J 22	Mixer Drive Home (Init)

Table 29-18: MagWashCU Board

29.9.2 Level Detection Board

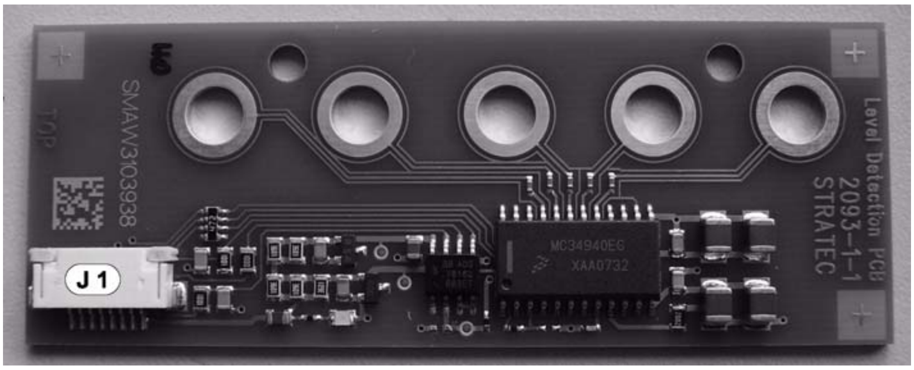

Figure 29-20: LEVEL DETECTION Board

Slot No.	Description
J 1	LEVEL DETECTION Board

Table 29-19: LEVEL DETECTION Board

29.10 Luminometer

29.10.1 Measurement CU Board

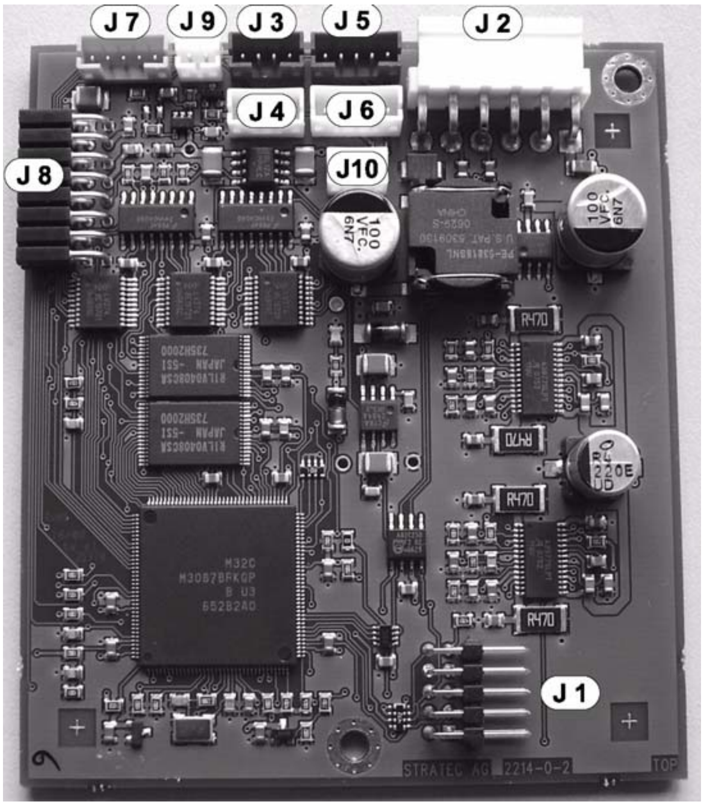

Figure 29-21: Measurement CU Board

Slot No.	Description
J 1	Download Monitor
J 2	Main PWR/CAN
J 3	Spindle Mtr
J 4	Lift Mtr.
J 5	Spindle Home
J 6	Lift Home
J 7	Ext.
J 8	PMT Modul
J 9	LED
J 10	Fan

Table 29-20: Measurement CU Board

29.10.2 HS PMT Module Board

Slot No.	Description
J 1	Main

Table 29-21: HS PMT Module Board

29.11 Incubator

29.11.1 Incubator CU Board

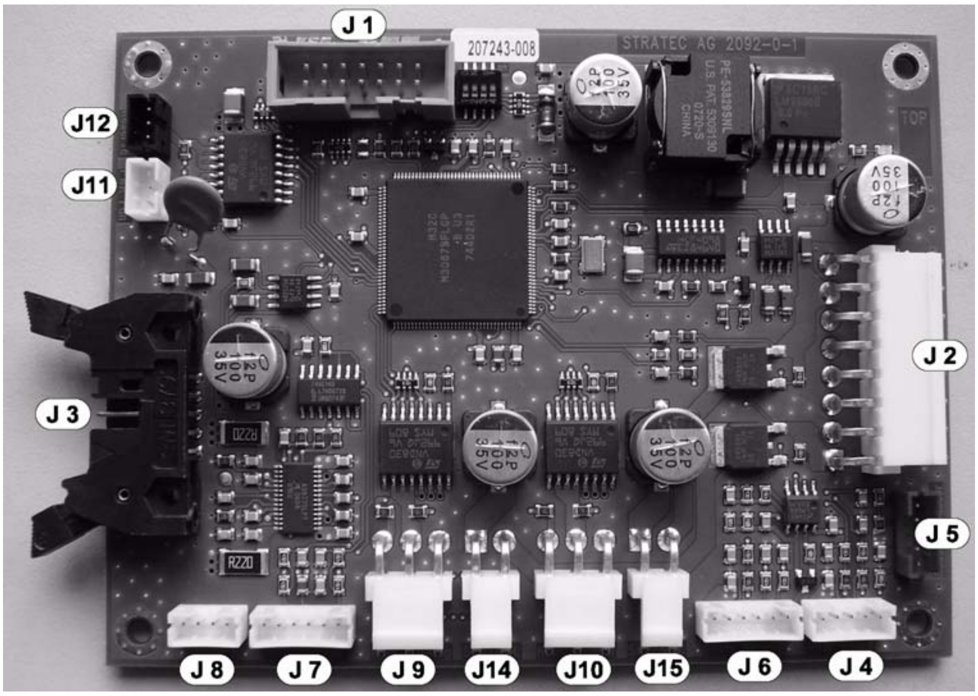

Figure 29-22: Incubator CU Board

Slot No.	Description
J 1	Download
J 2	Power-Can
J 3	Fluorometer
J 4	Door Board
J 5	Reserve Sensor
J 6	Axis Board
J 7	Encoder MTR
J 8	Carousel Motor
J 9	Heat Housing
J 10	Heat Bottom
J 11	PWR Fan
J 12	Reserve
J 14	Bimetall Switch_H
J 15	Bimetall Switch_B

Table 29-22: Incubator CU Board

29.11.2 Incubator Axis Board

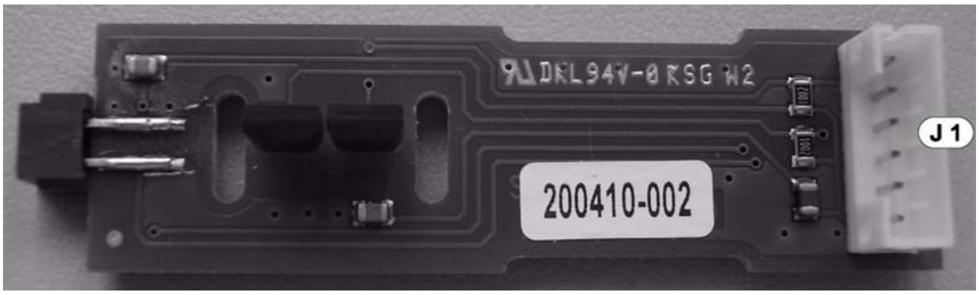

Figure 29-23: Incubator Axis Board

Slot No.	Description
J 1	To Incubator CU Board

Table 29-23: Incubator Axis Board

29.11.3 Incubator Door Board

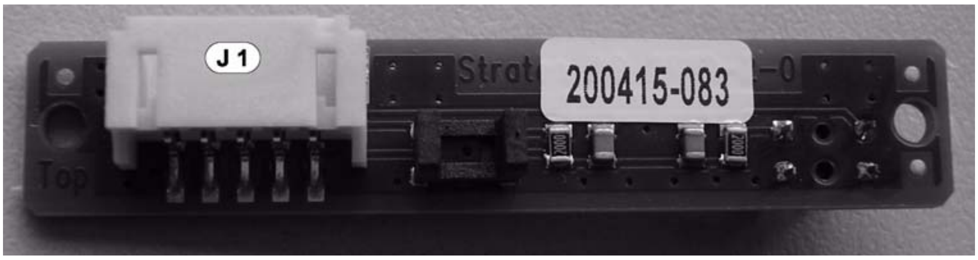

Figure 29-24: Incubator Door Board

Slot No.	Description
J 1	To Incubator CU Board

Table 29-24: Incubator Door Board

29.12 Input Queue

29.12.1 Input Queue CU Board

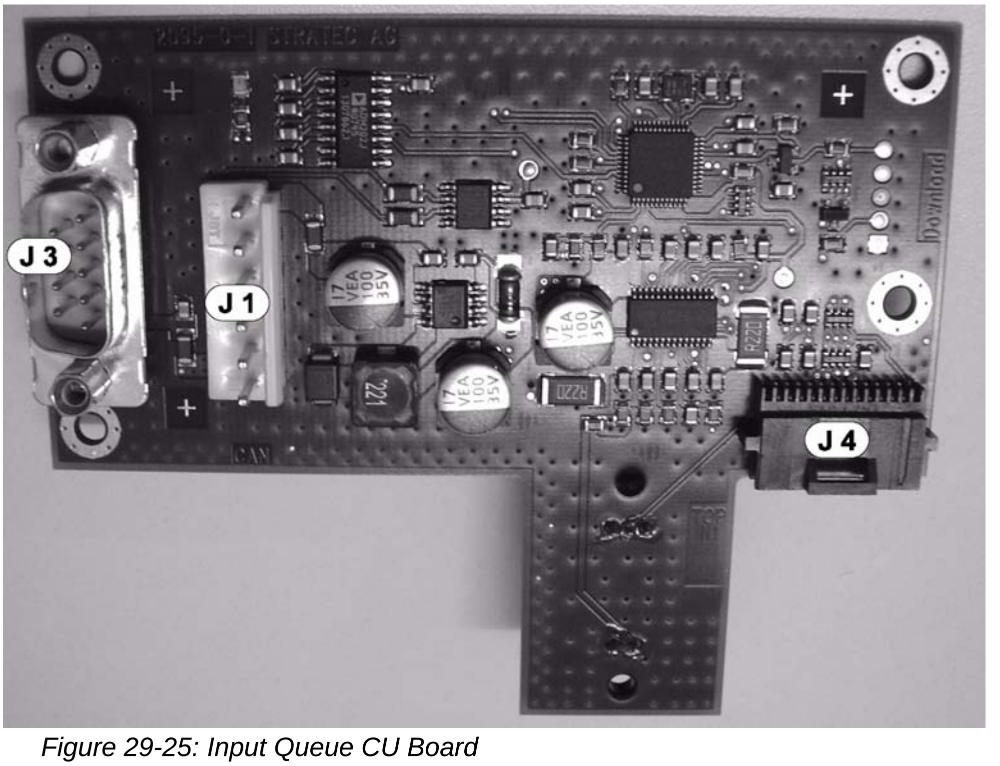

Figure 29-25: Input Queue CU Board

Slot No.	Description
J 1	Power CAN
J 2	Download
J 3	Barcode Scanner
J 4	Packing Drive Sensors + Motor

Table 29-25: Input Queue CU Board

29.12.2 Input Queue Sledge CU Board

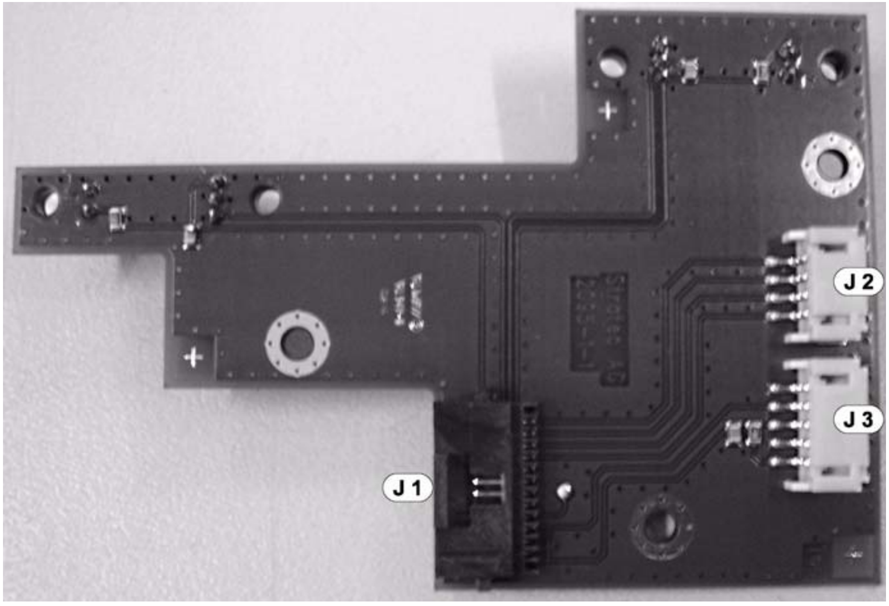

Figure 29-26: Input Queue Sledge CU Board

Slot No.	Description
J 1	To Input Queue CU Board
J 2	MTR Packing Drive
J 3	Packing Drive Rear Sensor

Table 29-26: Input Queue Sledge CU Board

Light Barrier No.	Description
IC 1	MTU PACKING-SENSOR
IC 2	PACKING DRIVE AHEAD-SENSOR

Table 29-27: Sensors, Input Queue Sledge CU Board

29.13 Output Queue

29.13.1 Output Queue CU Board

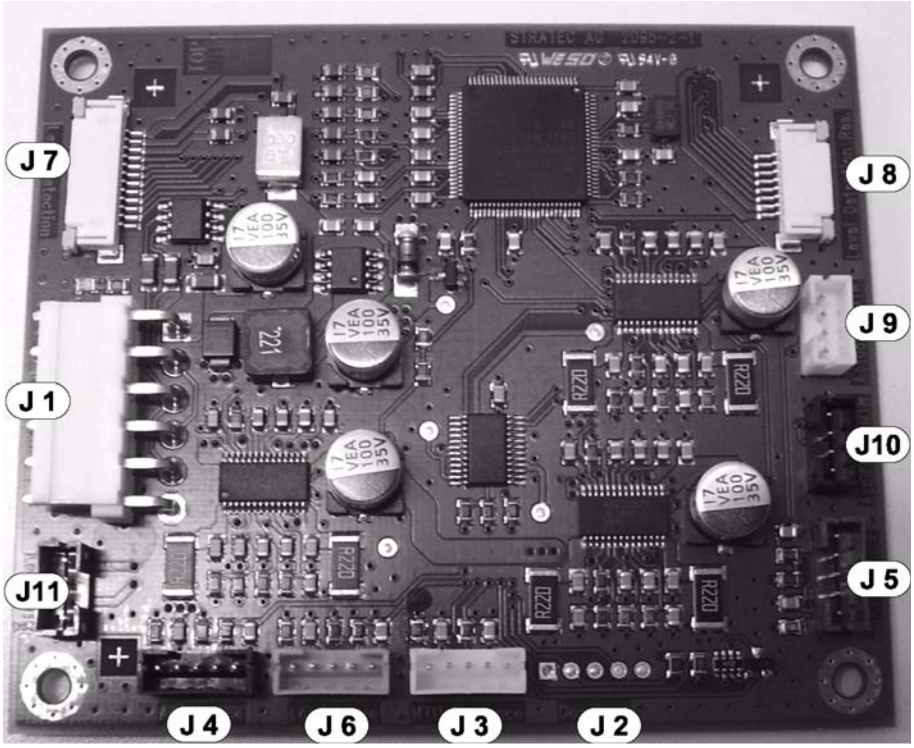

Figure 29-27: Output Queue CU Board

Slot No.	Description
J 1	Power-CAN
J 2	Download
J 3	MTU Presence
J 4	Actuator Home (Active Home)
J 5	Aspirator Home
J 6	Load Home
J 7	Level Detection + Sensors
J 8	Level Detection Reserve
J 9	Actuator Motor
J 10	Aspirator Motor
J 11	Load Motor

Table 29-28: Output Queue CU Board

29.13.2 Dispense Verification Board

Slot No.	Description
J 1	Output Queue CU Board
J 11	MTU Present

Table 29-29: Dispense Verification Board

29.14 Cooling Unit

29.14.1 Liquid CU Board

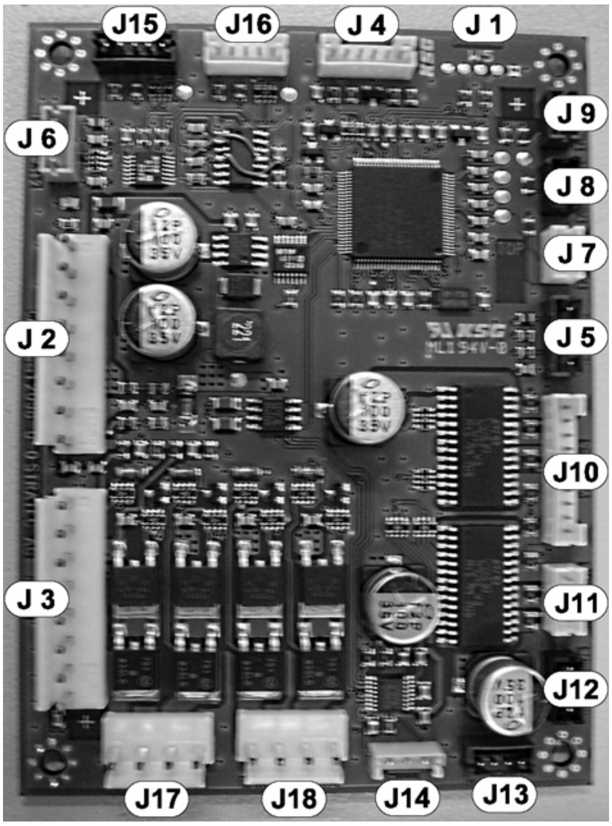

Figure 29-28: Liquid CU Board

Slot No.	Description
J 1	Download
J 2	CAN / Power 1
J 3	Power 2
J 4	Vacuum Pump ctrl
J 5	Pressure Sensor
J 6	Fload Switch 1+2
J 7	Temp. Cooling Modul
J 8	Temp. Ambient
J 9	Temp. Chiller Block
J 10	Valve 1-5
J 11	Res.1 Valve
J 12	Res.2 Valve
J 13	PWR Fan
J 14	Bimor Pump (DC Motor)
J 15	Bubble Sensor 1
J 16	Bubble Sensor 2
J 17	Peltier Modules 1
J 18	Peltier Modules 2

Table 29-30: Liquid CU Board